Harsh Desai

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Mechanical engineer with hands-on experience in scientific instrumentation, electromechanical test stands, vacuum/HV systems, plasma diagnostics, DAQ automation, and mechanical design; experienced in taking experimental hardware from setup and integration through testing and troubleshooting.

EDUCATION

University of Michigan — **Ann Arbor, MI, USA**
Master of Engineering, Space Systems Engineering | GPA: 3.79/4.0 — Aug 2024–Dec 2025

Coursework: Plasma & Fields Instrumentation; Plasma Measurement Techniques; Control for Aerospace Vehicles; Space Systems Design & Mgmt.

Vellore Institute of Technolo
Bachelor of Technology, Mec

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Fusion 360, Autodesk Inventor, AutoCAD, ANSYS Mechanical/Fluent
Instrumentation & Diagnostics: Langmuir probe, emissive probe, RPA, CEM/ESA calibration, SIMION, SRIM, Amptek DPPMCA
DAQ & Controls: Python, LabVIEW (CLAD prep), PyVISA/PyMeasure, Keithley DAQ/SMU, TDK Lambda PSUs, GPIB, serial, HV switching
Digital Readout: ADC/FPGA workflow, Zmod ADC 1410, Zynq-7000, AMD Vivado, MATLAB HDL Coder
Lab & Documentation: Vacuum chamber operation (CTI Hi-Vac / roughing to ∼1 mTorr), 1.3 kV test stands; Microsoft Word, Excel, PowerPoint; technical test plans and reports

ENGINEERING & INSTRUMENTATION EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan — **Ann Arbor, MI, USA**
Research Assistant, LVACCS Instrumentation & Test-Stand Characterization — Feb 2026–Present
- Built and operated an instrumented 1.3 kV HV test workflow with discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data capture
- Developed a Python/PyVISA + Tkinter operator GUI to automate ignition sequences, mass-flow control, and remote PSU coordination for high-throughput test-stand operation
- Operated Langmuir, emissive, and retarding-potential probes during vacuum-chamber experiments, performing bias sweeps and interpreting I–V / plasma-potential-related measurements across regolith-lofting test states
- Configured Keithley DAQ and TDK Lambda HV sequencing for hollow-cathode discharge testing (1.3 A constant-current, 1300 V headroom, xenon ignition near 500 V / 50 SCCM) on a CTI Hi-Vac / roughing-pumped chamber

Solar and Heliospheric Research Group, University of Michigan — **Ann Arbor, MI, USA**
Student Research Assistant, SSD Instrumentation Readout Chain — Jan 2025–Dec 2025
- Developed and verified an ADC/FPGA readout architecture for a solid-state detector, sizing digitization requirements and evaluating a 14-bit, 125 MSPS Zmod ADC / Zynq-7000 path using MATLAB and Vivado

Space Physics Research Lab, University of Michigan — **Ann Arbor, MI, USA**
Summer Intern, TestBedz Spacecraft Qualification Platform — May 2025–Jul 2025
- Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility instrumentation limits across 6 test-type workflows
- Encoded hardware metadata, facility operating envelopes, and acceptance criteria into machine-readable compatibility checks and traceable test plans for repeatable qualification workflows

L3Harris and University of Michigan — **Ann Arbor, MI, USA**
Communications Lead, Uranian Orbiter and Probe Mission Study — Sep 2024–May 2025
- Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% ($450M) cost reduction through an orbiter-only configuration trade

MECHANICAL DESIGN & ENGINEERING PROJECTS

CANSAT 2022 (NASA & AAS) | **Ranked 7th worldwide out of 42 teams** — Aug 2021–Jul 2022
- Designed and built a 10 m unidirectional tether-deployment mechanism using a DC motor and custom worm-gear spool for non-backdrivable load transfer and controlled descent of a 2 mm braided-nylon payload line
- Integrated payload-release and camera-stabilization hardware—elastic-lid release with torsion-spring latch plus a dual-servo 2-axis gimbal—and iterated mechanical fit, assembly, and field checkout for fixed 45° imaging

Space Instrumentation Calibration & Ion-Optics Series — Jul 2025–Dec 2025
Graduate Course: SPACE 571 | Supervisor: Prof. Stefano Livi
- Calibrated Channel Electron Multiplier (CEM) response and the charge-injection → CSA/shaper → MCA signal chain, extracting gain, centroid, FWHM, and bias-dependent count response
- Calibrated Cylindrical Electrostatic Analyzer (ESA) performance from lab energy-angle scans and SIMION/SRIM ion-optics models, quantifying transmission, angular acceptance, and energy resolution
- Modeled SIMION dual-Einzel beam expansion and Bessel-box filter response, and used SRIM TRANSMIT to estimate carbon-foil energy loss / straggling against TOF-derived exit energies

Spaceport America Cup / IREC | **Placed 23rd globally and 5th in Asia-Pacific** — Apr 2020–Jul 2021
- Launched a Mach 0.9 solid-motor sounding rocket to a 10,000 ft target altitude with the competition team
- Developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction and design iteration

CANSAT 2021 (NASA & AAS) | **Placed 13th globally and 7th in Asia-Pacific** — Aug 2020–Jul 2021
- Built MATLAB telemetry / ground-control software for dual maple-seed payload deployment monitoring, and modeled mono-wing descent with Blade Element Theory plus transient CFD